

An Adaptive Fast Algorithm for Periodic Coulomb Sums in Arbitrary Unit Cells

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Periodic Coulomb sums

- Electrostatics, molecular dynamics, materials: N point charges q_j at $\mathbf{y}_j$ in a unit cell $\mathcal{C}$ with *periodic boundary conditions*.
- The periodic potential is an infinite lattice sum

$$\Phi(\mathbf{x}) = \sum_{\mathbf{n} \in \mathbb{Z}^d} \sum_{j=1}^N G(\mathbf{x}, \mathbf{y}_j + L\mathbf{n}) q_j, \quad \hat{G}(\boldsymbol{\xi}) = \frac{1}{|\boldsymbol{\xi}|^2},$$

- $L = [\mathbf{a}_1, \dots, \mathbf{a}_d]$ is an *arbitrary* lattice — oblique in 2D, triclinic in 3D.
- Only *conditionally convergent*: charge neutrality $\sum_j q_j = 0$, tin-foil ($\mathbf{k} = \mathbf{0}$) convention.

Two features that make it hard

arbitrary cell shapes + highly *nonuniform* charge distributions

Two classical families — and a gap

Fast Ewald

- split kernel: smooth far-field by FFT + direct near-field.
- $O(N \log N)$ for (nearly) *uniform* charges (Liang et al., 2026).
- ✗ one global Ewald parameter — cannot localize; clustered charges push the near-field toward $O(N^2)$.

Fast multipole method

- fully adaptive — handles nonuniformity natively.
- cubic PBC classical (Greengard and Rokhlin, 1987).
- ✗ arbitrary cells only recently (Pei et al., 2023), and via $2d$ directional plane-wave expansions.

The gap

A *fully adaptive*, $O(N)$ solver for *arbitrary cells* and highly nonuniform charges.

This work: periodic DMK

Extend the **dual-space multilevel kernel-splitting** (DMK) framework (Jiang and Greengard, 2025), a fully adaptive $O(N)$ free-space convolution scheme, to periodic boundary conditions in arbitrary unit cells.

2D, oblique cells

10–26×

faster than the periodic FMM
on highly nonuniform charges

3D, triclinic cells

≈ **free-space DMK**

same speed as the non-periodic
baseline, edge ratios up to ~ 17

- One reciprocal-lattice Fourier series, *not* $2d$ directional expansions.
- Free parameter r_c *unifies* the FMM and fast Ewald summation.
- Open source: PeriodicDMK (C++ core with Julia interface).

Free-space DMK 1: split the kernel by length scale

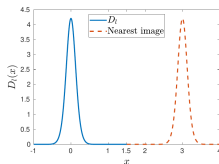
Direct evaluation $\Phi(\mathbf{x}_i) = \sum_j G(\mathbf{x}_i, \mathbf{y}_j) q_j$ costs $O(N^2)$. DMK (Jiang and Greengard, 2025) instead splits $G = \frac{1}{|\mathbf{x}-\mathbf{y}|}$ with a smooth function ϕ ($\phi(0) = 0$, $\phi(t) = 1$ for $t \geq 1$) at halving scales $h_\ell = h_0/2^\ell$ ($r = |\mathbf{x} - \mathbf{y}|$):

$$W_0(r) = \phi(r/h_0) G(r),$$

$$D_\ell(r) = [\phi(r/h_{\ell+1}) - \phi(r/h_\ell)] G(r),$$

$$R_{\mathcal{L}}(r) = [1 - \phi(r/h_{\mathcal{L}})] G(r).$$

- W_0 : smooth, $= G$ for $r \geq h_0$ — global tail (Fourier).
- D_ℓ : supported only in $r \leq h_\ell$.
- $R_{\mathcal{L}}$: short residual, summed directly.

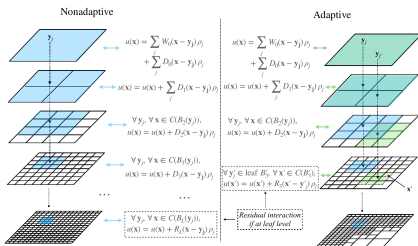


$\phi = \text{erf}$ is classical Ewald; DMK uses a prolate (PSWF) window.

D_ℓ is a localized bump.

Free-space DMK 2: one kernel per tree level

Sort the sources into an *adaptive* 2^d -tree (fine where charges cluster) and apply each kernel at its scale: W_0 at the root (global, once), D_ℓ at level ℓ (touching neighbors), R_ℓ at the leaves (direct).



W_0 needs only a short, N -independent Fourier transform; each finer D_ℓ couples only a box's $\leq 3^d$ neighbors. Two sweeps — *upward* (merge proxy charges, child $\rightarrow$ parent) and *downward* (translate between neighbors, parent $\rightarrow$ child) — touch each point $O(1)$ times $\Rightarrow O(N)$, fully adaptive.

From free space to periodic: periodize term by term

Periodize each component, $F^{(p)}(\mathbf{x}, \mathbf{y}) = \sum_{\mathbf{n}} F(\mathbf{x} - \mathbf{y} - L\mathbf{n})$:

$$G^{(p)} = W_0^{(p)} + \sum_{\ell=0}^{\mathcal{L}-1} D_{\ell}^{(p)} + R_{\mathcal{L}}^{(p)}.$$

Near field — local

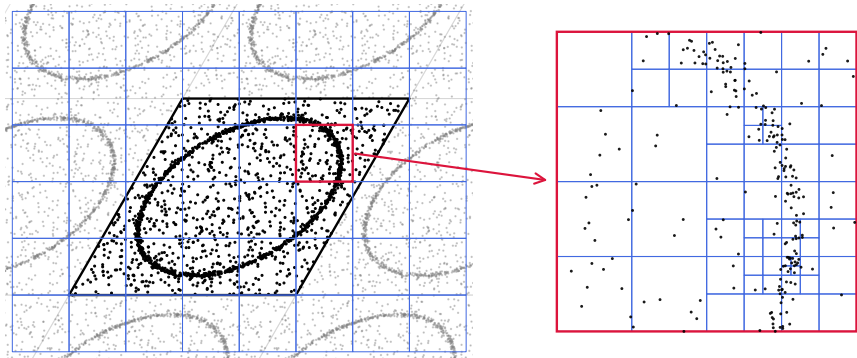
$D_{\ell}^{(p)}, R_{\mathcal{L}}^{(p)}$ are *absolutely* convergent:
only finitely many nearby images
contribute.

Far field — global

Only $W_0^{(p)}$ stays conditionally
convergent $\Rightarrow$ one Fourier series on the
reciprocal lattice.

$$\Phi(\mathbf{x}_i) = \underbrace{\sum_j W_0^{(p)}(\mathbf{x}_i, \mathbf{y}_j) q_j}_{\Phi_{\text{far}}} + \underbrace{\sum_j \left[\sum_{\ell} D_{\ell}^{(p)} + R_{\mathcal{L}}^{(p)} \right] q_j}_{\Phi_{\text{near}}}.$$

Geometry: multi-cube root + image halo



Left: embed $\mathcal{C}$ in a grid of r_c -cubes ($m_1 \times \cdots \times m_d$); add a one-cube halo of *image* sources $\rightarrow$ extended domain $\mathcal{E}$.

Right: a 2:1-balanced adaptive tree across the multi-cube root, refining with the local density.

Near field = free-space DMK on $\mathcal{E}$

- Run the free-space DMK on the *2:1 balance forest* $\mathcal{E}$ (originals + images), with only the root-level W_0 step *omitted*.
- Compact support $\Rightarrow$ this reproduces the periodized near-field *exactly*: every contributing image lies in $\mathcal{E}$.

Cost stays linear

Number of sources in $\mathcal{E}$ is linear to N .

- No new machinery — the periodic near-field is just a slightly modified free-space DMK call.
- Cubic cells are even simpler: virtual neighbors, no image particles (cf Klinteberg et al., 2026).

Far field = a single Fourier series

Poisson summation, with $\mathbf{k}_m = 2\pi L^{-T} \mathbf{m}$:

$$W_0^{(p)}(\mathbf{x}, \mathbf{y}) = \frac{1}{V} \sum_{\mathbf{m} \neq \mathbf{0}} \widehat{W}_0(|\mathbf{k}_m|) e^{i\mathbf{k}_m \cdot (\mathbf{x} - \mathbf{y})}, \quad V = |\det L|.$$

Effectively finite: $\widehat{W}_0$ is supported on $|\mathbf{k}_m| \lesssim c/r_c$.

Evaluate by the five-step fast-Ewald procedure (Liang et al., 2026) $\approx$ two NUFFT's (Barnett et al., 2019):

type-1 NUFFT $\rightarrow$ diagonal scaling by $\frac{1}{V} \widehat{W}_0(|\mathbf{k}_m|) \rightarrow$ type-2 NUFFT.

- **No upsampling** needed (post-scaling coefficients already decay) — saves a factor 2^d in the FFT.
- $\mathbf{m} = \mathbf{0}$ is dropped by charge neutrality.

The algorithm in five stages



- Stages 2–3 are the existing free-space DMK, run on $\mathcal{E}$.
- Stage 4 is the only periodic-specific work — two NUFFTs.
- Self-correction removes the spurious $W_0(\mathbf{0}) q_j$ at coincident source–target pairs.

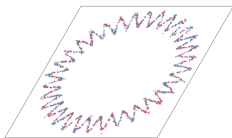
Complexity

Total cost

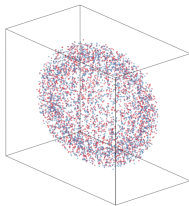
$$O(N + N_F \log N_F), \quad N_F \approx \frac{(c/r_c)^d V}{(2\pi)^d}.$$

- N_F (active Fourier modes) is fixed by the *cell shape* and precision — *independent of N* .
- $\Rightarrow O(N)$ at fixed cell shape; the near-field DMK dominates in practice (60–90% of the runtime).
- Anisotropy enters only as a constant overhead: $O(1)$ isotropic, $O(A \log A)$ rod, $O(A^2 \log A)$ slab, $A = L_{\max}/L_{\min}$.

Numerical results: highly nonuniform charges



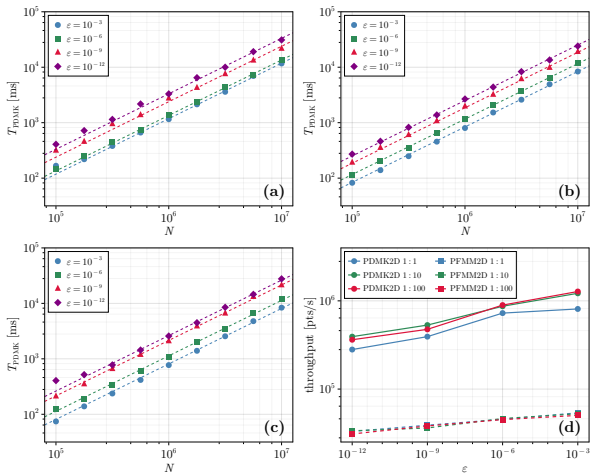
(a)



(b)

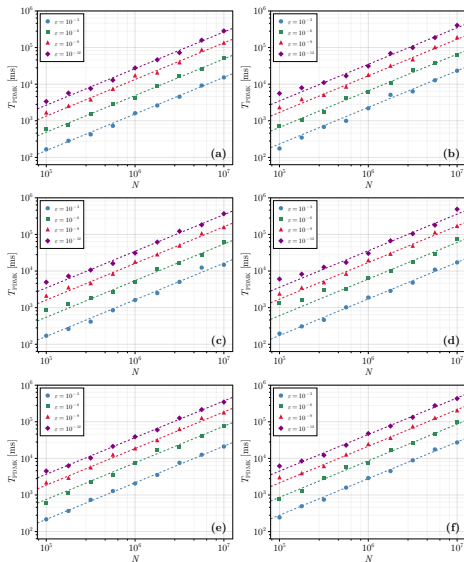
- 32-lobe perturbed ring (2D, top) / wavy spherical shell (3D, bottom).
- Local density varies $\sim 10\times$ over short scales — where uniform-grid Ewald struggles.
- 2D: oblique $\pi/3$ cells, aspect $A \in \{1, 10, 100\}$.
- 3D: rectangular & triclinic, regimes iso / rod / slab (edge ratios up to ~ 17).
- Single thread, double precision; N up to 10^7 , ε down to 10^{-12} .

2D: 10–26 $\times$ over the periodic FMM



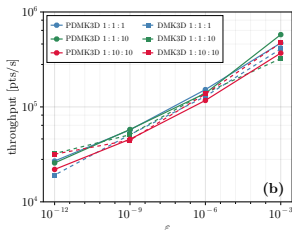
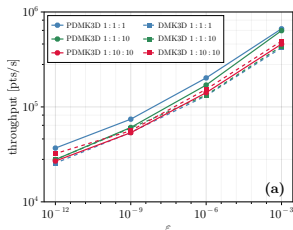
- (a–c): linear in N at every precision and aspect ratio.
- (d) throughput: **PDMK** (solid) sits 10–26 $\times$ above **PFMM** (dashed).
- Margin grows with aspect ratio, peaking near $A = 10$ – 100 .

3D: linear scaling on six geometries



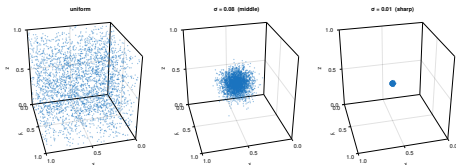
- Rows: iso / rod / slab regimes.
- Left: rectangular cell; right: triclinic counterpart.
- Linear in N everywhere; triclinic shear inflates runtime by only $\sim 1.1\text{--}1.5\times$, not the slope.

3D: as fast as free-space DMK

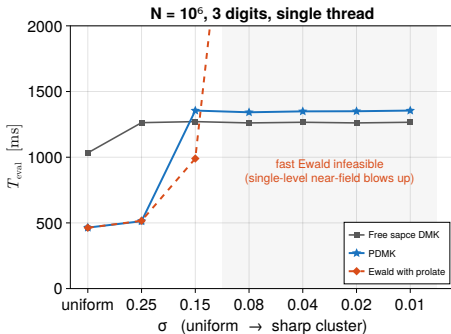


- **PDMK** (solid) tracks the *free-space DMK* (dashed) on the same sources.
- (a) rectangular, (b) triclinic.
- Ratio $0.6\text{--}1.8\times$ — periodicity is *nearly free*.

One more thing: r_c unifies fast Ewald and the DMK



one cloud, swept uniform $\rightarrow$ sharp spike



- **Fast Ewald** — best when uniform; single-level near field *blows up* as charges cluster.
- **Free-space DMK** — adaptive: flat in σ , yet pays that cost even when uniform.
- **PDMK** — follows the *cheaper* curve: Ewald when uniform, DMK when clustered.

r_c interpolates: small $\rightarrow$ Ewald, large $\rightarrow$ FMM.

Take-home

- **Periodic DMK:** a fully adaptive, $O(N)$ solver for periodic Coulomb sums in *arbitrary* unit cells (oblique 2D, triclinic 3D).
- Near field = free-space DMK on a multi-cube root + image halo; far field = *one* reciprocal-lattice Fourier series via two NUFFT's.
- r_c unifies the FMM and fast Ewald in a single implementation.
- 10–26 $\times$ over the periodic FMM in 2D; $\approx$ free-space DMK in 3D.

What's next

- Other kernels (DMK is largely kernel-agnostic)
- mixed / slab boundary conditions
- GPU & MPI

Open source

PeriodicDMK

- C++ core with a Julia interface.
- Built on the DMK software library + FINUFFT.
- Periodic *and* quasi-periodic (Bloch) Laplace sums, arbitrary cells, 2D & 3D.

Reproducible benchmarks
for every figure in the paper.

github.com/xuanzhaogao/PeriodicDMK

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Flatiron Institute

Thank you!

Questions?

Appendix: Periodic DMK algorithm

Input: Lattice L , sources $\{(\mathbf{y}_j, q_j)\}$, targets $\{\mathbf{x}_i\}$, precision ε .

- 1: *Preprocess.* Orient L upper-triangular; choose $r_c = L_{\min}/\kappa$, leaf capacity n_s ; compute root grid $m_i = \lceil w_i/r_c \rceil$; populate $\mathcal{E}$ with image charges.
- 2: *Tree.* Build the adaptive tree on $\mathcal{E}$ with $(m_1 + 2) \times \cdots \times (m_d + 2)$ root cubes of side r_c ; patch level-0 colleagues; enforce 2:1 balance across root cubes.
- 3: *Near field.* Run free-space DMK on $\mathcal{E}$ with the root W_0 step omitted $\Rightarrow G - W_0$.
- 4: *Far field.* Five-step NUFFT for $W_0^{(p)}$ on fractional coordinates $\mathbf{v}_j = L^{-1}\mathbf{y}_j$, $\mathbf{u}_i = L^{-1}\mathbf{x}_i$; drop $\mathbf{m} = \mathbf{0}$.
- 5: *Assemble.* $\Phi = \Phi_{\text{near}} + \Phi_{\text{far}}$; subtract $W_0(\mathbf{0}) q_j$ at coincident pairs.
- 6: **return** $\Phi(\mathbf{x}_i)$, $i = 1, \dots, M$.

Appendix: quasi-periodic (Bloch) conditions

$\Phi(\mathbf{x} + L\mathbf{n}) = e^{i\beta \cdot L\mathbf{n}}\Phi(\mathbf{x})$ for a Bloch vector β . Three localized modifications:

- 1 **Image charges** acquire the phase $q_j^{(n)} = q_j e^{i\beta \cdot L\mathbf{n}}$ in the near-field DMK.
- 2 **Far-field series** is evaluated at *shifted* modes $|\mathbf{k}_m + \beta|$ — *all* modes retained, charge neutrality no longer required:

$$W_0^{(p)} = \frac{1}{V} \sum_{\mathbf{m}} \widehat{W}_0(|\mathbf{k}_m + \beta|) e^{i(\mathbf{k}_m + \beta) \cdot (\mathbf{x} - \mathbf{y})}.$$

- 3 **Five-step** gains source pre-modulation $e^{-i\beta \cdot \mathbf{y}}$ and target post-modulation $e^{i\beta \cdot \mathbf{x}}$.

No other changes — the periodic case is $\beta = \mathbf{0}$.

Appendix: Fourier grid & self-correction

- **Mode grid.** The active modes occupy an ellipsoid in $\mathbf{m}$ -space; the smallest enclosing rectangular grid is

$$M_i = 2 \left\lceil \frac{c |\mathbf{a}_i|}{2\pi r_c} \right\rceil, \quad i = 1, \dots, d.$$

Rectangular-vs-ellipsoid overhead is $4/\pi$ (2D), $6/\pi$ (3D) for orthogonal cells, $\times$ a Hadamard factor $\prod_i |\mathbf{a}_i|/V$ for skew.

- **Self-correction.** The NUFFT adds a spurious $W_0(\mathbf{0}) q_j$; in 3D

$$W_0(\mathbf{0}) = \frac{\psi_0^c(0)}{4\pi c_0 r_c}, \quad c_0 = \int_0^1 \psi_0^c(s) ds,$$

subtracted at each coincident source–target pair.

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